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Visual Perception and Recognition of Emotional Facial Expressions by Deaf Individuals

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Objective: Auditory deprivation is hypothesized to influence other sensory systems, including vision, but it remains unclear whether these changes enhance or impair visual function. Additionally, it is uncertain whether low- and high-level visual processing are affected equally. ***Method:*** This analytical cross-sectional observational study included 63 participants (32 deaf and 31 hearing). Low-level visual functions were assessed using visual acuity and visual field tests, while high-level visual function was evaluated through emotional recognition in faces using a Stroop paradigm with emotional conflict. Statistical analyses included the D'Agostino–Pearson test and Wilcoxon signed-rank tests, with significance set at $\alpha = .05$. ***Results:*** Low-level visual function assessments revealed no significant differences in visual acuity between deaf and hearing participants. However, deaf participants demonstrated larger visual fields compared to the hearing group. In high-level visual function, deaf participants exhibited greater ease in suppressing word reading during emotional face recognition in a Stroop task compared to hearing participants. ***Conclusions:*** Deafness appears to influence visual perception by enhancing low-level processes, such as peripheral vision expansion, and improving high-level processes, such as emotional expression recognition. These findings highlight the intricate ways sensory deprivation may reshape visual processing.

Public Significance Statement
The deafness impacts the perception of vision. The deafness is related to better recognition of emotional expressions.

Keywords: deafness, facial expression recognition, Stroop test, visual perception

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continued

Several studies have demonstrated that auditory deprivation can lead to notable perceptual modifications, including changes in visual function when compared to individuals with typical hearing (Grégoire et al., 2022). Among these changes, individuals with hearing loss have shown larger visual fields, faster detection of stimuli, and quicker localization of visual targets (Bavelier et al., 2006).

The structural and functional alterations observed in the auditory cortex of individuals and animals with auditory deprivation appear to underlie these perceptual changes. For example, the auditory cortex can become responsive to visual stimuli, suggesting cross-modal reorganization (Almeida et al., 2015; Finney et al., 2001; Lambertz et al., 2005). Additionally, plasticity in other cortical areas, such as the visual cortex, plays a significant role in modifying perception (Grégoire et al., 2022; Pénicaud et al., 2013).

The expansion of the peripheral visual field observed in deaf individuals is thought to be associated with both retinal adaptations and increased activation of brain regions sensitive to motion stimuli (Bosworth & Dobkins, 2002; Codina et al., 2011; Neville & Lawson, 1987). Interestingly, certain visual measures, such as contrast sensitivity, appear to remain comparable between deaf individuals and those with typical hearing, indicating that not all aspects of visual function are affected by auditory deprivation (Finney & Dobkins, 2001).

The classification of visual function into low-level and high-level processing is based on several criteria, primarily focusing on the type of visual information processed and the complexity of the tasks involved (Bennett et al., 2019; Colenbrander, 2005). This framework reflects a hierarchical model of visual perception, where low-level processing handles basic visual features, such as edges, colors, and motion, while high-level processing involves more complex tasks, such as interpretation, recognition, and meaning extraction (Hegdé, 2018). Although low-level processing forms the foundation for initial perception, high-level processing is critical for deriving meaning and context. Importantly, the relationship between these two levels is not strictly unidirectional; they influence each other through dynamic feedback mechanisms. These interactions enable enhanced scene interpretation, demonstrating the integrative nature of visual perception (Nicol et al., 2013; Smith & Rossit, 2018).

Although modifications in low-level visual functions have been consistently observed in deaf individuals, findings regarding high-level visual processing, such as emotional recognition, remain inconclusive. Research shows that deaf signers, who rely heavily on visual communication, often perform as well as or better than hearing nonsigners in tasks like face recognition and short-term memory involving facial stimuli (Bettger et al., 1997; McCullough & Emmorey, 1997; Stoll et al., 2018). These results suggest that reliance on visual cues during social interactions may enhance sensitivity to subtle facial features (Letourneau & Mitchell, 2011; Mitchell et al., 2013).

However, evidence on emotion recognition among deaf individuals is more mixed. Some studies report enhanced emotion recognition, particularly in children, potentially due to their dependence on facial cues for communication (Hao & Su, 2014; Serra et al., 2017). Conversely, other studies highlight deficits in emotion recognition, often linked to delayed linguistic development and limited access to verbal descriptions of emotions (Dyck & Denver, 2003; Sidera et al., 2017). For instance, Sidera et al. (2017) found that

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deaf children underperformed compared to hearing peers in identifying basic emotions like happiness and sadness. Among adults, findings are sparse, but Letourneau and Mitchell (2011) demonstrated that deaf adults struggle with recognizing emotions when only partial facial information is available, emphasizing the importance of integrating cues from the entire face.

These conflicting results have led to competing hypotheses about visual emotion recognition in deaf individuals. The enhancement hypothesis suggests that sensory compensation following auditory deprivation heightens sensitivity to facial expressions, enabling superior emotion recognition through unique visual processing strategies (Arnold & Murray, 1998; Hao & Su, 2014). This aligns with evidence that deaf individuals excel in memory and recognition tasks and rely heavily on nonverbal communication, which demands the integration of various visual functions, such as acuity, motion detection, and texture perception (Arioli et al., 2023).

Conversely, the deficit hypothesis posits that the absence of auditory cues and limited exposure to emotional dialogue impede emotion recognition in deaf individuals (Sidera et al., 2017). Studies supporting this perspective report poorer facial emotion recognition among deaf and hard-of-hearing individuals compared to hearing controls (Dyck et al., 2004; Most & Michaelis, 2012; Wang et al., 2011). Emotion recognition is inherently a multimodal process, relying on visual, auditory, and spatiotemporal contextual information. The extent to which deaf individuals can compensate for the absence of auditory input in this process remains uncertain (Rodger et al., 2021).

In the present study, we compared low-level visual functions (visual acuity and visual field) and high-level visual function (facial emotion recognition) between deaf and hearing individuals. Low-level visual functions provide a sensory baseline for understanding differences between groups, while high-level visual function was assessed using an emotional Stroop task to evaluate the cognitive interference of emotional stimuli. We hypothesize that auditory deprivation induces changes in low-level visual functions, such as enhanced peripheral vision, which occur alongside alterations in high-level visual tasks, such as facial emotion recognition. The investigation of these functional changes in deaf people is important for understanding neuroplasticity and how the brain adapts to sensory deprivation and how it can influence different domains of their life.

Method

Ethical Considerations

This study was conducted according to the international ethical standards proposed by the Declaration of Helsinki and was approved by a local Ethics Committee (Report No. 3.597.811). The written informed consent was obtained from each participant.

Participants

This is an analytical cross-sectional observational study where 63 individuals were evaluated and divided into two groups: hearing and deaf. The hearing group consisted of 31 individuals (23 women and eight men; $M_{age} = 29 \pm 9.7$ years). The deaf group consisted of 32 individuals (13 women and 19 men; $M_{age} = 31 \pm 9$ years). In this group, 25 participants presented with congenital deafness and six with acquired hearing loss, having become deaf around 1 year of age. The reasons for congenital deafness in this sample, when described, were due to maternal lupus, toxoplasmosis, and an episode of eclampsia; however, most participants were not aware of the reason for congenital deafness. In cases of acquired deafness, all of which occurred during childhood, the causes were fever before the first year of age, meningitis, head trauma, and fall. None of the tested volunteers used hearing devices.

In both groups, all the volunteers were residents of the state of Maranhão, Brazil. The inclusion criteria for the deaf group were bilateral anacusis and adeptness with Língua Brasileira de Sinais (Brazilian Sign Language). The inclusion criterion for the hearing group was preserved hearing capacity.

The exclusion criteria in both groups included presentation of eye diseases such as glaucoma, cataract, and macular degeneration; infectious diseases such as toxoplasmosis, malaria, tuberculosis, and leprosy; chronic systemic diseases such as diabetes and hypertension; neurological diseases such as epilepsy; and being a carrier of neuropsychiatric diseases such as anxiety disorder and depression. They were interviewed about information regarding identification and contact, learning sign language (Língua Brasileira de

Sinais), personal and family health history, and life habits.

The preferred mode of communication for the deaf individuals was sign language. The reading task was assessed for both groups through a preevaluation and also during the Test for Recognition of Emotional Conflict From Facial Expression (TREFACE) 1, and both groups showed similar results.

Experimental Procedure

All participants then underwent audiological and neuro-visual evaluation tests. To classify the participants as deaf or hearing, the hearing tests performed were tonal, vocal, and impedance audiometrics (de Fonoaudiologia, 2020).

The low-level visual assessment aimed to estimate the visual acuity and the visual field perimetry. The visual acuity was evaluated using the Freiburg Visual Acuity Test (Bach, 2025). For that, we tested with a 15″ LCD display at 4 m, a luminance of 100 cd/m^2 at a contrast of 95% in a dark room. A Landolt C was presented on the screen, and the C-gap orientation was randomly changed every trial in four alternatives (top, bottom, right, left). The forced-choice of the participant fed an adaptive-staircase procedure, the best-Parameter Estimation by Sequential Testing algorithm (Lieberman & Pentland, 1982), that controlled the size of the optotype and calculated the visual spatial resolution (visual acuity), in decimal notation, of the participant. The visual field perimetry was estimated using the Förster perimeter. The Förster perimeter includes a semicircular arc positioned abroad that can be manually rotated around its axis. The participant had to maintain the fixation at a fixation point at the center of the arc, while the experimenter moved a movable white dot along the arc from the periphery toward the arc's center. The participant's task was to indicate the moment when the white dot was visually detected. The procedure was conducted at 40 rotational angles of the arc, and the experimenter recorded the angle at which the white dot was detected for later perimetric area quantification in AUTOCAD software.

The TREFACE was used to analyze facial discrimination ability (Prada et al., 2022). The test was applied in three steps. The first step involved the presentation of a stimulus where a face was presented paired with a word and consisted of 70 attempts to match the face–word stimulus. This initial step was considered a condition so that the tested person could become familiar with the stimulus. For the second and third steps, the stimuli were accompanied by the word in the center of the face (on the centerline between the eyes and nose) without affecting the central details of the character; both steps presented congruent (face and word matched) and incongruent (face and word did not match) stimuli. In the second step, the person tested was instructed to correctly read the word on the screen and ignore the face. In the third step, the person tested was instructed to verbally indicate one in five emotional expressions (fear, disgust, anger, joy, and surprise) of the faces and ignore the word presented in the stimulus (Prada et al., 2022).

For all presentations, the stimulus size was 5.5 × 8 degrees of visual angle with a white background. The participant was invited to sit in a comfortable chair that was placed 80 cm away from a 17″ LCD screen and was positioned at a 90° angle. During the TREFACE task, the participant had to respond to the instructions provided for the different test steps. A Língua Brasileira de Sinais interpreter was hired to explain each step of the test and to decode the response.

A total of 210 stimuli were presented, 70 for each stage, which were either congruent (the emotion and presented word corresponded) or incongruent (the emotion and the presented word did not correspond).

For the analysis of the answers, we counted the number of correct answers (answer matched), errors (answer differed), and omissions (not answered; Prada et al., 2022).

Statistics

For statistical analysis, the data were evaluated for normality using the D'Agostino–Pearson test. We observed all parameters were nonparametric, and to compare the results obtained from both groups, we used the Wilcoxon signed-rank test. In all cases, a value of $\alpha = .05$ was considered significant. We used the Jamovi software to proceed with the statistical analysis.

Results

Low-Level Vision Assessment

The mean visual acuities measured for the deaf group and hearing group were 0.91 ± 0.43 and

1.01 ± 0.35 decimal units. There was no statistically significant difference in visual acuity between the deaf and hearing groups (p = .3992). The visual field area of the deaf group was 35.93 ± 13.4 cm^2, which was significantly larger (p = .001) than the visual field area of the hearing group, which was 29.43 ± 9.8 cm^2. In the visual field evaluation, the deaf group presented a greater visual field area than the hearing group.

High-Level Vision Assessment

No participant had an error in the initial step of TREFACE. It was observed that there was no significant difference in performance (number of correct responses, errors, and omissions) between the hearing and deaf groups in the second step of TREFACE (Table 1, $p > .05$). However, for the third step of TREFACE, significant performance differences were observed between the groups (Table 2, $p < .05$). The deaf group had higher correct responses and lower errors and omissions compared to the hearing group, indicating a lesser emotional conflict between faces and words to recognize faces.

Discussion

The results obtained in this study showed that there was no difference in visual acuity between the deaf and hearing participants and that there was a presence of a larger visual field in deaf participants than in people with normal hearing. Additionally, we also observed that deaf participants had less conflict between words and faces when the attentional focus was on facial expression recognition.

The literature shows that the sensory loss of hearing could have a compensatory counterpart in other sensory systems, such as an improvement in some aspects of visual function (Codina et al., 2011). This statement assumes that deaf individuals would compensate for hearing loss with improved visual perception of the environment (Codina et al., 2011). Controversial results about spatial vision from deaf people have been reported. Some studies have observed no significant differences between the visual acuity of deaf and hearing people (Brancaleone et al., 2025; Finney & Dobkins, 2001), while other investigations found loss of high spatial frequency contrast sensitivity (Khorrami-Nejad et al., 2018). The analysis of the peripheral visual field in this study showed that deaf individuals recognized events occurring in the periphery of the visual field more accurately than the hearing group. One of the possible explanations may be the understanding that visuospatial materiality is important for the processes of sign language acquisition in deaf individuals (Lopes & Leite, 2011). In this sense, visual perception can be improved in its attentional aspect as sign language shapes its spatial references (Lopes & Leite, 2011). People who are or become deaf early in life are taught to be visually attentive (Marin & Góes, 2006).

Our initial hypothesis regarding functional changes in low- and high-level visual functions was partially confirmed. Deaf participants demonstrated an increased visual field, which coincided with better performance in facial emotion

Table 1
Performance of the Participants in the Second Step of TREFACE Considering All Presentations, Congruent Stimuli, and Incongruent Stimuli

Parameter	Hearing group	Deaf group	p
All stimuli			
Number of correct responses	68 (7)	69 (7)	.98[a]
Number of errors and omissions	2 (7)	1 (7)	.98[a]
Congruent stimuli			
Number of correct responses	35 (1)	35 (6.25)	.07[a]
Number of errors and omissions	0 (1)	0 (6.25)	.075[a]
Incongruent stimuli			
Number of correct responses	33 (7)	35 (4.25)	.47[a]
Number of errors and omissions	2 (7)	0 (3.25)	.34[a]

Note. The values are represented as median (interquartile range). TREFACE = Test for Recognition of Emotional Conflict From Facial Expression.
[a] Wilcoxon signed-rank test.

Table 2
Performance of the Participants in the Third Step of TREFACE Considering All Presentations, Congruent Stimuli, and Incongruent Stimuli

Parameter	Hearing group	Deaf group	*p*
All stimuli			
Number of correct responses	51 (18.5)	64.5 (17)	.002[a]
Number of errors and omissions	19 (18.5)	5.5 (17)	.002[a]
Congruent stimuli			
Number of correct responses	28 (9)	32 (4.25)	.001[a]
Number of errors and omissions	7 (9)	3 (4.25)	.001[a]
Incongruent stimuli			
Number of correct responses	25 (11.5)	31 (10.25)	.004[a]
Number of errors and omissions	10 (11.5)	4 (10.25)	.004[a]

Note. The values are represented as median (interquartile range). TREFACE = Test for Recognition of Emotional Conflict From Facial Expression.
[a] Wilcoxon signed-rank test.

recognition during a Stroop test. These findings align with previous studies suggesting that visual field restriction significantly impacts facial expression recognition by increasing inspection time, the number of fixations, and fixation duration (Urtado et al., 2024). This connection highlights the interplay between peripheral visual enhancements and the ability to process emotional cues from facial expressions.

TREFACE assesses the ability to recognize facial expressions with emotional content in another human being under conditions of conflict or not between what is read and what is visualized. Our findings indicate that in the step in which the focus of attention was the written word and not the facial expression (Step 2), the deaf group and hearing group had similar performances. In contrast, in the step in which the attention was on the facial expression instead of the written word (Step 3), the deaf group showed greater ease in performing the task, with fewer errors than the hearing group, nevertheless the congruence and incongruence between faces and words.

The findings here suggest that the better performance in Step 3 of deaf people compared to hearing people represents a reduction in the emotional conflict between recognizing faces and the meaning of expressions. This finding suggests that deaf people's strategies for recognizing faces in the face of other stimuli may be different from those of hearing people. This gives particularities to their experience and interaction with the environment, whether linguistic, discursive, or interpretive (Marin & Góes, 2006). If we consider that sign language organizes visuospatial materiality, we can assume that the visual tasks that involve both the physiological framework of seeing and understanding facial expressions of emotions that the deaf individuals acquire demonstrate that the deaf are visual, and not because of amplified visual sense or development of focused visual culture due to lack of hearing.

In this direction, a study (McCullough & Emmorey, 1997) focused on the visual-cognitive performance of deaf individuals who showed improved abilities along the dimensions of facial processing. In general, it is believed that the improvement of such visual skills is also a result of experience in sign language, as the understanding and acquisition of the language strongly depend on such visuospatial skills. Claudino et al. (2020) considered that facial expression in sign language did not necessarily favor face recognition because they are possibly two different neuropsychological aspects, and deaf individuals pay attention to the face because they provide critical intonational and linguistic information during communication. The emotional recognition task used in the present study is related to the ability to recognize emotions in a very specific situation, where there is (or is not) a conflict between the expression and a word. Therefore, it may be more connected to abilities beyond just recognizing emotions (e.g., a lower tendency to focus on words or a greater tendency to focus on faces, rather than an improved ability to recognize faces). This is especially relevant considering that no differences were found in Step 1, but differences did emerge in Step 3, both in congruent and incongruent tasks.

The present study has a limitation regarding the unequal proportion of male and female participants between the groups, with the control group having a higher proportion of women compared to the deaf group. The literature indicates that women generally perform better than men in emotion recognition tasks (Peng et al., 2023; Rafiee & Schacht, 2023). However, this difference in gender distribution is unlikely to have compromised the results qualitatively, as the deaf group outperformed the hearing group in emotion recognition, despite having fewer women. Our sample consisted of deaf signers, which limited the generalizability of our findings, particularly given that our method of emotion recognition relied on the conflict between recognizing emotions and interpreting word meanings. Both groups were literate in Portuguese, and no specific assessment of reading performance was conducted. Since all participants had at least 14 years of formal education, we consider that any differences in reading performance likely had minimal influence on the results of the emotional recognition task. Future investigations could incorporate additional measurements, such as reaction times, to enhance the scope of our interpretations.

These results may contribute to a better understanding of the visual processing strategies of deaf people and open up possibilities for further manipulation of the stimuli involved in conflicting visuospatial tasks.

Conclusion

Overall, the results of this study indicate that deaf individuals have better peripheral vision than hearing individuals. In the recognition of facial expressions with emotional content using the Stroop paradigm, the deaf individuals showed better performance than hearing individuals. These data suggest that deaf individuals are better adapted to the visual world, and factors such as visual attention should be relevant in the recognition strategy of facial expressions.

References

Almeida, J., He, D., Chen, Q., Mahon, B. Z., Zhang, F., Gonçalves, Ó. F., Fang, F., & Bi, Y. (2015). Decoding visual location from neural patterns in the auditory cortex of the congenitally deaf. *Psychological Science*, *26*(11), 1771–1782. https://doi.org/10.1177/0956797615598970

Arioli, M., Segatta, C., Papagno, C., Tettamanti, M., & Cattaneo, Z. (2023). Social perception in deaf individuals: A meta-analysis of neuroimaging studies. *Human Brain Mapping*, *44*(16), 5402–5415. https://doi.org/10.1002/hbm.26444

Arnold, P., & Murray, C. (1998). Memory for faces and objects by deaf and hearing signers and hearing nonsigners. *Journal of Psycholinguistic Research*, *27*(4), 481–497. https://doi.org/10.1023/A:1023277220438

Bach, M. (2025). Freiburg vision test (FrACT): Optimal number of trials? *Graefe's Archive for Clinical and Experimental Ophthalmology*, *263*(2), 437–442. https://doi.org/10.1007/s00417-024-06638-z

Bavelier, D., Dye, M. W., & Hauser, P. C. (2006). Do deaf individuals see better? *Trends in Cognitive Sciences*, *10*(11), 512–518. https://doi.org/10.1016/j.tics.2006.09.006

Bennett, C. R., Bex, P. J., Bauer, C. M., & Merabet, L. B. (2019). The assessment of visual function and functional vision. *Seminars in Pediatric Neurology*, *31*, 30–40. https://doi.org/10.1016/j.spen.2019.05.006

Bettger, J., Emmorey, K., McCullough, S., & Bellugi, U. (1997). Enhanced facial discrimination: Effects of experience with American sign language. *The Journal of Deaf Studies and Deaf Education*, *2*(4), 223–233. https://doi.org/10.1093/oxfordjournals.deafed.a014328

Bosworth, R. G., & Dobkins, K. R. (2002). The effects of spatial attention on motion processing in deaf signers, hearing signers, and hearing nonsigners. *Brain and Cognition*, *49*(1), 152–169. https://doi.org/10.1006/brcg.2001.1497

Brancaleone, M. P., Boucher, L. C., Yang, J., Merfeld, D., & Onate, J. A. (2025). Comparing dynamic visual acuity between athletes who are deaf or hard-of-hearing and athletes who are hearing. *Journal of American College Health*, *73*(1), 27–30. https://doi.org/10.1080/07448481.2023.2198018

Claudino, R. G., da Costa, I. F., Nascimento, I. N. D. A., & Torro, N. (2020). Use of sign language does not favor recognition of static and dynamic emotional faces in deaf people. *Psychology & Neuroscience*, *13*(4), 531–538. https://doi.org/10.1037/pne0000238

Codina, C., Pascalis, O., Mody, C., Toomey, P., Rose, J., Gummer, L., & Buckley, D. (2011). Visual advantage in deaf adults linked to retinal changes. *PLOS ONE*, *6*(6), Article e20417. https://doi.org/10.1371/journal.pone.0020417

Colenbrander, A. (2005). Visual functions and functional vision. *International Congress Series*, *1282*, 482–486. https://doi.org/10.1016/j.ics.2005.05.002

de Fonoaudiologia, C. F. (2020). *Guia de Orientação na Avaliação Audiológica* [Guidance guide for audiological assessment]. Conselho Federal de Fonoaudiologia.

Dyck, M. J., & Denver, E. (2003). Can the emotion recognition ability of deaf children be enhanced? A pilot study. *The Journal of Deaf Studies and Deaf Education*, *8*(3), 348–356. https://doi.org/10.1093/deafed/eng019

Dyck, M. J., Farrugia, C., Shochet, I. M., & Holmes-Brown, M. (2004). Emotion recognition/understanding ability in hearing or vision-impaired children: Do sounds, sights, or words make the difference? *Journal of Child Psychology and Psychiatry*, *45*(4), 789–800. https://doi.org/10.1111/j.1469-7610.2004.00272.x

Finney, E. M., & Dobkins, K. R. (2001). Visual contrast sensitivity in deaf versus hearing populations: Exploring the perceptual consequences of auditory deprivation and experience with a visual language. *Cognitive Brain Research*, *11*(1), 171–183. https://doi.org/10.1016/S0926-6410(00)00082-3

Finney, E. M., Fine, I., & Dobkins, K. R. (2001). Visual stimuli activate auditory cortex in the deaf. *Nature Neuroscience*, *4*(12), 1171–1173. https://doi.org/10.1038/nn763

Grégoire, A., Deggouj, N., Dricot, L., Decat, M., & Kupers, R. (2022). Brain morphological modifications in congenital and acquired auditory deprivation: A systematic review and coordinate-based meta-analysis. *Frontiers in Neuroscience*, *16*, Article 850245. https://doi.org/10.3389/fnins.2022.850245

Hao, J., & Su, Y. (2014). Deaf children's use of clear visual cues in mindreading. *Research in Developmental Disabilities*, *35*(11), 2849–2857. https://doi.org/10.1016/j.ridd.2014.07.034

Hegdé, J. (2018). Neural mechanisms of high-level vision. *Comprehensive Physiology*, *8*(3), 903–953. https://doi.org/10.1002/cphy.c160035

Khorrami-Nejad, M., Heravian, J., Askarizadeh, F., & Sobhani-Rad, D. (2018). Contrast sensitivity abnormalities in deaf individuals. *Journal of Ophthalmic & Vision Research*, *13*(2), 153–157. https://doi.org/10.4103/jovr.jovr_218_16

Lambertz, N., Gizewski, E. R., de Greiff, A., & Forsting, M. (2005). Cross-modal plasticity in deaf subjects dependent on the extent of hearing loss. *Cognitive Brain Research*, *25*(3), 884–890. https://doi.org/10.1016/j.cogbrainres.2005.09.010

Letourneau, S. M., & Mitchell, T. V. (2011). Gaze patterns during identity and emotion judgments in hearing adults and deaf users of American sign language. *Perception*, *40*(5), 563–575. https://doi.org/10.1068/p6858

Lieberman, H. R., & Pentland, A. P. (1982). Microcomputer based estimation of psychophysical thresholds: The best PEST. *Behavior Research Methods and Instrumentation*, *14*(1), 21–25. https://doi.org/10.3758/BF03202110

Lopes, M. A. D. C., & Leite, L. P. (2011). Concepções de surdez: A visão do surdo que se comunica em língua de sinais [Conceptions of deafness: The perspective of the deaf person who communicates through sign language]. *Revista Brasileira de Educação Especial*, *17*(2), 305–320. https://doi.org/10.1590/S1413-65382011000200009

Marin, C. R., & Góes, M. C. R. D. (2006). A experiência de pessoas surdas em esferas de atividade do cotidiano [The experience of deaf individuals in everyday spheres of activity]. *Cadernos CEDES*, *26*(69), 231–249. https://doi.org/10.1590/S0101-32622006000200007

McCullough, S., & Emmorey, K. (1997). Face processing by deaf ASL signers: Evidence for expertise in distinguished local features. *The Journal of Deaf Studies and Deaf Education*, *2*(4), 212–222. https://doi.org/10.1093/oxfordjournals.deafed.a014327

Mitchell, T. V., Letourneau, S. M., & Maslin, M. C. (2013). Behavioral and neural evidence of increased attention to the bottom half of the face in deaf signers. *Restorative Neurology and Neuroscience*, *31*(2), 125–139. https://doi.org/10.3233/RNN-120233

Most, T., & Michaelis, H. (2012). Auditory, visual, and auditory–visual perceptions of emotions by young children with hearing loss versus children with normal hearing. *Journal of Speech, Language, and Hearing Research*, *55*(4), 1148–1162. https://doi.org/10.1044/1092-4388(2011/11-0060)

Neville, H. J., & Lawson, D. (1987). Attention to central and peripheral visual space in a movement detection task: An event-related potential and behavioral study. II. Congenitally deaf adults. *Brain Research*, *405*(2), 268–283. https://doi.org/10.1016/0006-8993(87)90296-4

Nicol, J. R., Perrotta, S., Caliciuri, S., & Wachowiak, M. P. (2013). Emotion-specific modulation of early visual perception. *Cognition and Emotion*, *27*(8), 1478–1485. https://doi.org/10.1080/02699931.2013.793654

Peng, D., Zheng, W. L., Liu, L., Jiang, W. B., Li, Z., Lu, Y., & Lu, B. L. (2023). Identifying sex differences in EEG-based emotion recognition using graph convolutional network with attention mechanism. *Journal of Neural Engineering*, *20*(6). https://doi.org/10.1088/1741-2552/ad085a

Pénicaud, S., Klein, D., Zatorre, R. J., Chen, J.-K., Witcher, P., Hyde, K., & Mayberry, R. I. (2013). Structural brain changes linked to delayed first language acquisition in congenitally deaf individuals. *NeuroImage*, *66*, 42–49. https://doi.org/10.1016/j.neuroimage.2012.09.076

Prada, E., Satler, C., Tavares, M. C. H., Garcia, A., Martinez, L., Alves, C., Lacerda, E., & Tomaz, C. (2022). TREFACE: A new computerized test of emotional Stroop with facial expressions. *Journal of Behavioral and Brain Science*, *12*(7), 342–358. https://doi.org/10.4236/jbbs.2022.127020

Rafiee, Y., & Schacht, A. (2023). Sex differences in emotion recognition: Investigating the moderating effects of stimulus features. *Cognition and Emotion*, *37*(5), 863–873. https://doi.org/10.1080/02699931.2023.2222579

Rodger, H., Lao, J., Stoll, C., Richoz, A. R., Pascalis, O., Dye, M., & Caldara, R. (2021). The recognition of facial expressions of emotion in deaf and hearing individuals. *Heliyon*, *7*(5), Article e07018. https://doi.org/10.1016/j.heliyon.2021.e07018

Serra, A., Spinato, G., Cocuzza, S., Licciardello, L., Pavone, P., & Maiolino, L. (2017). Adaptive psychological structure in childhood hearing impairment: Audiological correlations. *Acta Otorhinolaryngologica Italica*, *37*(3), 175–179. https://doi.org/10.14639/0392-100X-1291

Sidera, F., Amadó, A., & Martínez, L. (2017). Influences on facial emotion recognition in deaf children. *The Journal of Deaf Studies and Deaf Education*, *22*(2), 164–177. https://doi.org/10.1093/deafed/enw072

Smith, F. W., & Rossit, S. (2018). Identifying and detecting facial expressions of emotion in peripheral vision. *PLOS ONE*, *13*(5), Article e0197160. https://doi.org/10.1371/journal.pone.0197160

Stoll, C., Palluel-Germain, R., Caldara, R., Lao, J., Dye, M. W. G., Aptel, F., & Pascalis, O. (2018). Face recognition is shaped by the use of sign language. *The Journal of Deaf Studies and Deaf Education*, *23*(1), 62–70. https://doi.org/10.1093/deafed/enx034

Urtado, M. B., Rodrigues, R. D., & Fukusima, S. S. (2024). Visual field restriction in the recognition of basic facial expressions: A combined eye tracking and gaze contingency study. *Behavioral Sciences*, *14*(5), Article 355. https://doi.org/10.3390/bs14050355

Wang, Y., Su, Y., Fang, P., & Zhou, Q. (2011). Facial expression recognition: Can preschoolers with cochlear implants and hearing aids catch it? *Research in Developmental Disabilities*, *32*(6), 2583–2588. https://doi.org/10.1016/j.ridd.2011.06.019

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